

MUHAMMAD TUAHA BIN ZAHEEN

Electrical Engineer | Power Systems, Embedded Systems & Digital Twins | AI & Cloud-Enabled Engineering Solutions
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PROFESSIONAL SUMMARY

PEC-Registered Electrical Engineer and Researcher with experience in electrical and electronic systems, power systems, embedded systems, instrumentation, control, and testing. Skilled in circuit design, PCB design, system integration, troubleshooting, and technical analysis. Experienced in developing IoT, digital twin, and cloud-enabled engineering solutions, with a strong focus on R&D, problem-solving, system development, and engineering innovation.

CORE COMPETENCIES & TECHNICAL SKILLS

Leadership & Professional Skills: Technical Leadership, Team Collaboration, Cross-Functional Coordination, Engineering Problem-Solving, Analytical Thinking, Critical Thinking, Decision-Making, Project Coordination, Innovation, Technical Communication

Electrical & Electronics Engineering: Power Systems & Smart Distribution, Analog & Digital Electronics, Instrumentation & Testing, Embedded Systems (Raspberry Pi, ESP, PIC), UART, SPI, I2C, CAN, Ethernet, ESD & EMC Compliance, PCB Design

AI & Intelligent Systems: LLMs, Azure OpenAI, Machine Learning, AI Automation, Digital Twins, NLP, Prompt Engineering, Embeddings, Semantic Search, AI Product Design, Computer Vision, OpenCV, Transformer Models, Scikit-learn

Cloud, Backend & Infrastructure: Microsoft Azure, Azure IoT Hub, Azure Digital Twins, Azure Functions, AWS, Tuya IoT Cloud, Cloud Microservices, REST APIs, SOAP Web Services, Redis, Nginx

Programming & Software Development: Python, C#, C++, Kotlin, Swift, Assembly, HDL/FPGA, FastAPI, Streamlit, API Development & Testing, Microservices Architecture

Quality Assurance & Release Management: QA Testing, Regression Testing, Functional Testing, REST & SOAP API Testing, User Acceptance Testing (UAT), Beta Testing, Defect Tracking, Release Validation, Performance Testing

Engineering & Simulation Tools: MATLAB, Simulink, Proteus, SolidWorks, AutoCAD (2D/3D), PCB Design

Product & Design: Product Management, AI Product Strategy, Figma, UX/UI Design, Agile/Scrum, SDLC, User Story Creation, Sprint Planning, Technical Documentation

Tools & Platforms: Git, GitHub, Jira, Confluence, SAP, Notion, Postman

PROFESSIONAL EXPERIENCE

Science & Engineering Associate

Jul 2025 – Present

United Bank Limited (UBL) RHQ, Innovation & Creative Technology Digital Lab — Islamabad, Pakistan

- Led microservices architecture projects for digital banking platforms across Pakistan and UAE, including a WiFi detection mechanism for the mobile banking app.
- Implemented Dynatrace SDK monitoring, D-Engage in-app notifications, and M-Tag enablement to improve application performance and customer engagement.
- Designed and managed research, design, and development of mobile banking solutions across iOS and Android within an AI-driven Digital Innovation Lab.
- Built QA regression testing systems for iOS/Android builds and supported UX/UI enhancements in Figma, improving customer journey flows and release readiness.
- Supported R&D on emerging digital technologies and automation, contributing to backend support and change-management initiatives under Agile/SDLC.

System Engineering Intern (EMC and ESD Lab)

Aug 2024 – Sep 2024

Rohde & Schwarz GmbH & Co. KG — Islamabad, Pakistan

- Operated spectrum analyzers, oscilloscopes, VNAs, and signal generators for RF and electronics hardware validation in an ESD-compliant laboratory.
- Supported embedded firmware troubleshooting and safety-compliance documentation; explored SAP system architecture for enterprise workflows.

Electrical Engineer Intern

Aug 2022 – Sep 2022

Pakistan Telecommunication Company Limited (PTCL) — Network & Power Systems, Islamabad, Pakistan

- Monitored optical transport networks and telecom power systems across 20+ nodes, maintaining approximately 99.9% uptime.

KEY PROJECTS

TwinWatt — Digital Twin-Enabled Smart Meter via Microsoft Azure

Jul 2024 – Jun 2025

- Developed a Digital Twin and AI-based smart meter using Raspberry Pi, Azure IoT Hub, and Azure Digital Twins for real-time energy monitoring and fault detection, aligned with UN SDG-7 and SDG-9.
- Awarded 2nd Position (Runner-up), Undergraduate Final Year Project Competition 2025, Bahria University; published in IEEE ICET 2025.

Matchify AI — Resume Screening System (NLP & LLM-Based Matching)

Aug 2025 – Oct 2025

- Built a full-stack candidate-matching system (FastAPI, Streamlit) using Sentence-Transformers and Llama3-70B for semantic scoring and structured candidate insights, with exploration of vision-language models for multimodal resume understanding.

Shuttle Bot — Autonomous Shuttlecock Collection Robot

Sep 2024 – Dec 2024

- Designed a computer-vision-guided robot (Raspberry Pi, Python, OpenCV) for real-time detection, pickup, and deposition, with mechanical and electronic design simulated in TinkerCAD.

WiFi-Enhanced Robotic Platform — Line Tracking & Obstacle Evasion

Jan 2023 – May 2023

- Built a PIC/ESP-based autonomous platform with IR and ultrasonic sensors and WiFi telemetry for real-time monitoring and control.

PUBLICATIONS

M. T. Bin Zaheen, R. S. Arshad, A. N. Ul Ain, A. N. Sarwar and M. H. D. Khan, "Digital Twin-Based Smart Metering Framework with Real-Time Monitoring via Microsoft Azure IoT," 2025 20th International Conference on Emerging Technologies (ICET), Peshawar, Pakistan, 2025, pp. 1-7, doi: 10.1109/ICET66147.2025.11321353.

EDUCATION

Bachelor of Science in Electrical (Electronics) Engineering

Sep 2021 – Jul 2025

Bahria University, School of Engineering and Applied Sciences — Islamabad, Pakistan

- CGPA: 3.34/4.0. Final Year Project: Digital Twin-Based Smart Meter Framework via Microsoft Azure (2nd Position, FYP Competition 2025; published in IEEE ICET 2025).

CERTIFICATIONS & AWARDS

- 5G Wireless Technology — Rohde & Schwarz
- Microsoft Azure AI Fundamentals & Generative AI Solutions — Microsoft
- Introduction to AI — IBM, Scrimba
- Cognitive Robotics — Alison
- Industry 4.0 & Industrial Automation — LinkedIn Learning
- 2nd Position, Undergraduate FYP Competition 2025
- Merit Scholarship, Fall 2021
- 1st Position, Minute to Win — NUST EME Olympiad 2023

LEADERSHIP & COMMUNITY

- Founder & President, Microsoft Learn Student Ambassadors (MLSA) — BSEAS Chapter, Bahria University (2024–2025): Partnered with Scrimba (Norway).
- Head of Operations, IEEE Student Branch, Bahria University (2024–2025): led outreach and digital-marketing strategy for campus-wide technical engagement.
- Lead – Outreach, Partnerships & Sponsorships, BUMUN Model United Nations (2023–2024).
- Marketing Lead, Google Developer Student Club (GDSC BUIC) (2023–2024).
- Vice President & Deputy Campus Director Hult Prize BUIC.
- Vice President & Deputy Campus Director Zindigi Prize BUIC.